Fiscal Year:	FY 2007	**Task Last Updated:**	FY 11/13/2007
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Project Title:	Apoptosis and Immune Homeostasis During Hindlimb Unloading		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	08854	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2005	**End Date:**	03/31/2009
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roberts, Arthur (University of Med & Dentistry of NJ) Das, Jyoti (Robert Wood Johnson Medical School) Carr, Jamie (University of Med & Dentistry of NJ)		
Grant/Contract No.:	NCC 9-58-IIH00405		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space flight is often associated with lymphopenia and reductions in the response to mitogens, and production of cytokines and antibodies. Factors such as microgravity, lack of load-bearing, psychological and physical stress, and irradiation are believed play important roles in these immune system effects, but the exact mechanisms are unknown. Ground-based animal models, such as hindlimb unloading (HU) in rodents, can simulate some of these factors. HU causes similar changes in the immune system as those resulting from space flight. Based on our preliminary results, we hypothesize that some of these changes are modulated through Fas, in a manner dependent on endogenous opioids, and that space radiation interacts synergistically to amplify these effects. We propose 3 specific aims: 1) Elucidate the role of Fas and endogenous opioids modulation of the immune system by HU and radiation; 2) Determine the effect of HU and radiation on lymphocyte dynamics and immune responses, and 3) because autoimmune disease, paradoxically, is often associated with stress and radiation exposure, we will determine whether HU and radiation promote autoimmunity. The overall goal of these studies is to determine the mechanisms underlying apoptosis-related immune modulation using ground-based models. PROGRESS 1. We have found that granzyme B (GzmB) plays critical role in regulating the number of Th2 cells. GzmB-deficient mice develop stronger Th2 type immune responses. Because GzmB expression can be inhibited by steroid hormones, and the Th2 response and corticosteroid levels are often elevated during stress, our finding may explain why astronauts have increased Th2 cytokines. We have found that mice subjected to HU also have greater Th2 responses. We will examine the role of GzmB in the immune response of mice subjected to HU. We also are collaborating with Satish Mehta and Duane Pierson of Johnson Space Center to examine the changes in GzmB in astronauts. 2. We continued to study the role of vitamin C in preventing HU-induced thymus involution. A manuscript describing our findings has been reviewed and resubmitted. We are currently collaborating with Ann Kennedy to examine the effect of vitamin C on radiation-induced changes in the immune system. 3. We continued our studies on the elevation in haptoglobin in HU-treated mice. We have found that serum haptoglobin levels correlate with stress levels. We have now extended our studies to astronaut serum; in collaboration with Duane Pierson and Satish Mehta, we have examined two batches of serum from 20 astronauts. We recently found a correlation between serum haptoglobin and viral reactivation status in astronauts and appropriate controls. We will examine additional samples to verify this correlation. Our ultimate goal here is to develop haptoglobin as a marker for evaluating the stress status of astronauts, as well as civilians. 4. We have recently examined the effects of adult stem cells on radiation-induced apoptosis of splenocytes and thymocytes in vitro and in vivo. Using bone marrow-derived mesenchymal stem cells, our experiments show that these cells help prevent radiation-induced apoptosis in the spleen and thymus. After confirming this result, we will begin to determine the molecular mechanism. Identification of the culpable molecules will allow development of countermeasures against radiation-induced lymphopenia. 5. We are examining the interesting communication between the immune and skeletal systems. Osteopontin (OPN), originally described as a bone matrix protein, has been shown to play a role in the immune system--as a cytokine and possibly a transcription factor. Since OPN and RANKL are closely related in the bone, we examined role of OPN in HU-induced lymphocyte reduction using OPN-knockout mice in collaboration with David Denhardt and Kathryn Wang at Rutgers University. We found that OPN-deficiency creates resistant to HU-induced thymocyte depletion. OPN seems to regulate the steroid response during HU. 6. In collaboration with Marcelo Vazquez, formerly of Brookhaven Laboratory, we investigated the long-term effect of particle radiation on immune cells. Six month after several low doses of 1.0 and 0.6 GeV/n 56Fe irradiation, most populations of immune cells examined showed few consistent changes. There was a moderate reduction in B cells numbers, however. We have applied for and received approval to conduct our own exposures at NSRL scheduled for April 10 of this year to examine the acute effects of low-dose 1.0 GeV/n 56Fe irradiation on lymphocytes in vitro and in vivo. FUTURE PLANS 1. Continue to examine the role of GzmB-regulated Th2 responses during HU treatment. We will work with Satish Mehta and Duane Pierson to examine the changes of GzmB in astronauts. 2. Complete the experiments on the role of CD4+CD25+ T regulatory cells in HU induced lymphocyte apoptosis and submit the manuscript. 3. Continue our studies with Duane Pierson and Satish Mehta examining the correlation between serum levels of haptoglobin and viral reaction status in astronauts to further confirm whether haptoglobin can be used as a diagnostic stress marker. 4. Continue our studies to identify cell survival/ radiation-protective molecules from mesenchymal stem cells. Since these cells have great potential in tissue repair, we will collaborate with Jerry Sonnenfeld to study the effects of mesenchymal stem cells on wound healing. 5. Complete the studies on the role of OPN in HU-induced thymocyte reduction. 6. Conduct preliminary our experiment on the acute effects of low-dose particle radiation on lymphocytes on April 10 at NSRL. This initial experiment should provide clues to the mechanisms involved. Based on the results of these experiments, we will propose new experiments to better define the dose effects and mechanisms, and examine the combined effects of HU and particle radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We hope to gain an understanding of the mechanisms by which HU affects immune homeostasis. With this information, it may be possible to devise dietary and lifestyle changes which reduce the deleterious effects of stress on the immune system. Such changes may lead to the improvement of public health. We also hope to identify food supplements, such as vitamin C and seaweed extracts that reduce the health impact of stress and radiation, bringing benefit to the general population, especially those who suffer from high levels of occupational or life stress or radiation. By determining the role of cellular apoptosis in mediating the deleterious effects of stress on the immune system we may be able to aid in the development of protocols for the maintenance of immune tolerance for clinical use in organ transplant or autoimmune disease. We are studying the role of a specific subset of T lymphocytes called CD4+CD25+ regulatory T cells which seem to be involved in the deleterious effect of stress on the immune system. Learning how these cell exert their effects may lead to potential interventions to prevent the negative health impacts of autoimmunity, cancer, and other diseases that arise from an improperly regulated immune system. It is difficult to measure the level of stress that an individual experiences for the purpose of allowing appropriate intervention. Identification of particular serum

	proteins whose concentration is related to the stress level could serve as a biomarker for the assessment of stress levels for medical or occupational purposes.
Task Progress:	1. Elucidate the role of Fas and endogenous opioids in the modulation of the immune system by hindlimb unloading (HU) and radiation: We have found that the synergistic effect of radiation and HU in a mouse model is also exerted through Fas since anti-FasL blocks their combined effect. 2. Determine the effect of HU and radiation on lymphocyte dynamics and immune responses: Artificial depletion of Tregs by administration of antibody against CD25 prevented HU-induced lymphocyte reduction. Osteopontin (OPN)-deficient mice are resistant to HU-induced lymphocyte reduction, suggesting that controlling OPN activity may provide a countermeasure. Vitamin C and antioxidant-rich food supplements helped prevent HU-and radiation-induced lymphocyte loss. Seaweed polysaccharide extract can inhibit radiation-induced lymphocyte loss. In collaboration with Marcelo Vazquez, formerly of Brookhaven Laboratory, we investigated the long-term effect of particle radiation on immune cells. Six month after several low doses of 1.0 and 0.6 GeV/n 56Fe irradiation, most populations of immune cells showed few consistent changes. There was a moderate reduction in B cells numbers, however. These were very long time points with mice receiving very low doses. Therefore, new experiments are planned using shorter time points and higher doses. 3. Determine whether HU and radiation promote the development of autoimmunity: We have examined the effect of apoptotic and necrotic cells on immune response, and found that unlike apoptotic cells, necrotic cells stimulate immune responses. In early ongoing studies of the rates of development of autoimmunity following HU with and without concomitant irradiation, our findings as yet are inconclusive. 4. Additional studies not in the original aims: The degree of stress in an individual is difficult to quantify due to the lack of a specific marker. Using a proteomics approach, we found that transferrin and haptoglobin were consistently increased in the serum of mice following HU. To extend this finding to the astronaut population, we are collaborating with Drs. Duane Pierson and Satish Mehta of Johnson Space Center, who have provided serum samples from astronauts pre-and post-flight. We found increased haptoglobin in astronaut serum in comparison to a control group. Thus, serum haptoglobin levels may provide a biomarker to quantitatively measure stress levels in astronauts.
Bibliography Type:	Description: (Last Updated: 11/19/2010)
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